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# **1/4 BPS Black Hole Entropy from the Gravitational Path Integral in $\mathcal{N} = 4$ Supergravity.**

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# Contents

|          |                                                                           |          |
|----------|---------------------------------------------------------------------------|----------|
| <b>1</b> | <b>The Plan:</b>                                                          | <b>1</b> |
| 1.1      | Construct a Test to ensure Falsifiability . . . . .                       | 1        |
| 1.2      | The Approach . . . . .                                                    | 1        |
| <b>2</b> | <b>Introduction</b>                                                       | <b>2</b> |
| 2.1      | Falsifiability . . . . .                                                  | 2        |
| 2.2      | Dabholkar’s Localisation in $\mathcal{N} = 2$ supergravity . . . . .      | 3        |
| 2.3      | IMT2022 . . . . .                                                         | 3        |
| 2.4      | Lopes N4 GPI . . . . .                                                    | 3        |
| <b>3</b> | <b>The Renormalised Action</b>                                            | <b>4</b> |
| 3.1      | The One-Loop Determinant . . . . .                                        | 5        |
| 3.2      | The Instanton Contribution . . . . .                                      | 5        |
| 3.3      | The Contour of Integration for the Single-Centered Contribution . . . . . | 5        |
| 3.4      | The Single-Centered Contribution . . . . .                                | 6        |
| <b>4</b> | <b>Method</b>                                                             | <b>7</b> |
| 4.1      | Scalar Manifold . . . . .                                                 | 7        |
| <b>5</b> | <b>Definitions</b>                                                        | <b>8</b> |
| <b>6</b> | <b>app1</b>                                                               | <b>9</b> |

# 1 The Plan:

## 1.1 Construct a Test to ensure Falsifiability

- Compute  $d(m, n, \ell)$  from  $\frac{1}{\phi_{10}}$  for small charge by expanding an Igusa cusp for from its product formula.
- Numerically evaluate right-hand side of Thm 5.1 for the same charges summing over  $\gamma$  up the same cutoff.
  - (a) Compute eq (6.45) CNR (  $\Delta = m = \ell = 0$  case.) Thus for all gravity proposals there exists a check
  - (b)
    - Polar coefficients  $c_m^F(\tilde{n}, \tilde{l})$
    - Multiplier and Kloosterman sums  $\psi(\Gamma)_{\ell, \tilde{\ell}}$
    - Bessel's  $I_{\frac{23}{2}}$
    - Eichler Integral in term *III*

## 1.2 The Approach

- Field content and multiplets
- Localise
- One-loop determinant
- The measure
- Integration
- Sum over saddles

## 2 Introduction

We define the Helicity Supertrace by

$$B_n = \frac{1}{n!} \text{Tr} (-1)^{2h} (2h)^n \quad (2.1)$$

In  $\mathcal{N} = 4$  in 4 dimensions  $\exists$  16 supercharges, 4 of which are preserved. Thus, 12 are broken into 6 pairs ( $2^6 = 64$ ). Thus we should consider the index  $B_6$ .

If we want to obtain  $B_6$  from gravity it would be one asymptotically flat path integrals with several saddle classes.

$$-B_6(Q, P; \mu) = d_{\text{hair}} \cdot d_{\text{horizon}}(\Gamma) + \sum_{\Gamma_1 + \Gamma_2 = \Gamma} \Theta_\mu(\Gamma_1, \Gamma_2) (-1)^{\langle \Gamma_1, \Gamma_2 \rangle + 1} |\langle \Gamma_1, \Gamma_2 \rangle| d_{\frac{1}{2}}(\Gamma_1) d_{\frac{1}{2}}(\Gamma_2) \quad (2.2)$$

where  $\Gamma = (Q, P)$  is the charge vector living in the Narain lattice  $\Lambda_{6,22}$ , and we have the DSZ pairing  $\langle \Gamma_1, \Gamma_2 \rangle = Q_1 \cdot P_2 - Q_2 \cdot P_1$  where  $\cdot$  is the  $O(6, 22)$  invariant inner product,  $Q \cdot P \equiv Q^T L P$  where  $L$  is the signature of  $(6, 22)$ . The first term in (2.2) is the single centered contribution while the second is the multi-centered contribution.

Theorem 5.1's left side is the immortal (single-centred) degeneracy in the R-chamber, which coincides with a fixed-chamber coefficient only for  $n > m$ .

### 2.1 Falsifiability

$$\frac{1}{\phi_{10}(\rho, \sigma, \nu)} = q^{-1} s^{-1} r^{-1} \prod_{(k,l,j) > 0} \left(1 - q^k s^l r^j\right)^{-c(4kl-j^2)}$$

$$\sum_{N \geq 0} p^N \chi(\text{Sym}^N K3; \tau, z) = \prod_{k > 0, l \geq 0, j} \left(1 - p^k q^l y^j\right)^{-c(4kl-j^2)}.$$

The Fourier coefficients of the elliptic genus of  $K3$  obeys

$$c(4n - j^2) = 2\phi_{0,1}(\tau, z) \quad (2.3)$$

where

$$\phi_{0,1} = \phi_{-2,1}(1 + 12P) + 12A \quad A = \prod_{n \geq 1} \frac{(1 - q^n y)^2 (1 - q^n y^{-1})^2}{(1 - q^n)^4}$$

$$\wp / (2\pi i)^2 = \frac{1}{12} + \frac{y}{(1 - y)^2} + \underbrace{\sum_{n \geq 1} q^n \sum_{d|n} d \left( y^d - 2 + y^{-d} \right)}_P.$$

We can truncate  $\frac{1}{\phi_{10}}$  using  $(1 - x)^{-e} = \sum_t \binom{e+t-1}{t} x^t$

## 2.2 Dabholkar's Localisation in $\mathcal{N} = 2$ supergravity

Dabholkar says

$$Z = \int_{\mathcal{M}_Q} |Z_{\text{inst}}|^2 |Z_{1\text{-loop}}| d\mu_Q \exp \left[ -\pi \phi^I q_I + \mathcal{F}(\phi, p) \right] \quad (2.4)$$

"For example, with  $N = 4$  supersymmetry, in gauge theory both  $|Z_{\text{instanton}}|^2$  and  $Z_{\text{det}}$  equal unity. Similarly, it was found in [15] that very similar determinant factors for vector multiplets equal unity  $N = 4$  theories. One expects that this simplification will extend to the factors appearing in (5.52) around the localizing solution in  $N = 4$  theories."

## 2.3 IMT2022

"The technical steps necessary for performing the localization computation rely on several steps, none of which require working solely with the  $\mathcal{N} = 8$  theory, and should be applicable to theories with less supersymmetry. Theories of  $\mathcal{N} = 4$  supergravity obtained by a dimensional reduction of  $T^2 \times K3$ . One might hope to reproduce the exact degeneracy of 1/4 BPS and 1/2 BPS black holes. In the former case one has to remove the contribution of multi-center black holes from the index. The resulting index for single-center black holes is the Fourier coeicient of a mock Jacobi form [120] [Dabholkar:2012nd] whose Rademacher expansion is more complicated than the one for true Jacobi forms that we considered in this paper [121, 122] [Dabholkar:2012nd, LopesCardoso:2022hvc]  
pg 77

## 2.4 Lopes N4 GPI

The Holomorphic prepotential for  $\mathcal{N} = 4$  is given by

$$F(Y, \Upsilon) = F^{(0)}(Y) + 2i\omega(Y, \Upsilon)$$

$$F^{(0)}(Y) = -\frac{Y^1 Y^a \eta_{ab} Y^b}{Y^0} \quad \omega(Y, \Upsilon) = \frac{1}{256\pi} \Upsilon \log \eta^{24}(iS) \quad S = -i \frac{Y^1}{Y^0}.$$

where  $a, b = 2, \dots, 23$ ,  $Y^I$  are the off shell fields in the supergravity formalism before localisation. We have the Dedekind Eta function give by

$$\eta(\tau) = q^{\frac{1}{24}} \prod_{n=1}^{\infty} (1 - q^n) \quad (2.5)$$

Once we perform the localisation, we have  $Y^I = \frac{1}{2}(\phi^I + ip^I)$  and  $\Upsilon = -64$ .

They suggest a suitable measure to contain the  $\eta$  contribution from the Holomorphic prepotential ( eq 4.25)

$$\tilde{\mu}_1 = \quad (2.6)$$

### 3 The Renormalised Action

We are concerned with the contribution from the re-normalised action

$$\mathcal{S}_{\text{ren}} = \frac{\pi q_I \phi^I}{c} - \frac{4\pi}{c} \Im F \left( \frac{\phi^I + ip^I}{2} \right) \quad (3.1)$$

where we evaluate the prepotential on the localising manifold and  $I = 0, 1, \dots, 23$ . Thus we can examine the pre-potential

$$F \left( \frac{\phi^I + ip^I}{2} \right) = - \frac{(\phi^1 + ip^1)(\phi^a + ip^a)\eta_{ab}(\phi^b + ip^b)}{4(\phi^0 + ip^0)} - \frac{i}{2\pi} \log \eta^{24} \left( \frac{\phi^1 + ip^1}{\phi^0 + ip^0} \right).$$

We multiply by the conjugate of both terms over itself

$$F \left( \frac{\phi^I + ip^I}{2} \right) = - \frac{(\phi^1 + ip^1)(\phi^a + ip^a)\eta_{ab}(\phi^b + ip^b)(\phi^0 - ip^0)}{4[(\phi^0)^2 + (p^0)^2]} - \frac{i}{2\pi} \log \eta^{24} \left( \frac{(\phi^1 + ip^1)(\phi^0 - ip^0)}{(\phi^0)^2 + (p^0)^2} \right).$$

The imaginary part of the first term is given by

$$-\eta_{ab} \frac{-\phi^1 \phi^a \phi^b p^0 + \phi^1 \phi^a p^b \phi^0 + \phi^1 p^a \phi^b \phi^0 + \phi^1 p^a p^b p^0 + p^1 \phi^a \phi^b \phi^0 + p^1 \phi^a p^b p^0 + p^1 p^a \phi^b p^0 - p^1 p^a p^b \phi^0}{4[(\phi^0)^2 + (p^0)^2]}.$$

If we introduce the contractions  $A := \phi^a \eta_{ab} \phi^b - p^a \eta_{ab} p^b$ ,  $B := \phi^a \eta_{ab} p^b + p^a \eta_{ab} \phi^b$  we have

$$\Im F^{(0)} = - \frac{(\phi^1 \phi^0 + p^1 p^0) B + (p^1 \phi^0 - \phi^1 p^0) A}{4[(\phi^0)^2 + (p^0)^2]} \quad (3.2)$$

For the second term if we let  $z = \frac{(\phi^1 + ip^1)(\phi^0 - ip^0)}{(\phi^0)^2 + (p^0)^2}$ ,  $q = e^{2\pi i z}$  and note that  $\log z = \ln |z| + i \text{Arg} z$

$$-\frac{i}{2\pi} \log \eta^{24} \left( \frac{(\phi^1 + ip^1)(\phi^0 - ip^0)}{(\phi^0)^2 + (p^0)^2} \right) = -\frac{i}{2\pi} \left( \log \left[ q \prod_{n=1}^{\infty} (1 - q^n)^{24} \right] \right) \quad (3.3)$$

$$-\frac{i}{2\pi} \left( \log q + \log \left[ \prod_{n=1}^{\infty} (1 - q^n)^{24} \right] \right) = -\frac{i}{2\pi} \left[ 2\pi i z + 24 \sum_{n=1}^{\infty} \log (1 - q^n) \right].$$

So the second term is

$$z - \frac{12i}{\pi} \sum_{n=1}^{\infty} \log (1 - q^n) \quad (3.4)$$

Thus, the imaginary part of the second term is

$$\Im z - \frac{12}{\pi} \Re \sum_{n=1}^{\infty} \log (1 - q^n) = \Im z + \frac{12}{\pi} \Re \sum_{j=1}^{\infty} \sum_{n=1}^{\infty} \frac{(q^n)^j}{j} \quad (3.5)$$

Explicitly,

$$\frac{p^1\phi^0 - \phi^1p^0}{(\phi^0)^2 + (p^0)^2} - \frac{12}{\pi} \Re \sum_{n=1}^{\infty} \log(1 - q^n) \quad (3.6)$$

**XXX Check Commented part XXX**

We have that for large  $\Im z$ ,  $\log(1 - q^n) \rightarrow 0$

If we combine these contributions we have the form of the imaginary part of the Holomorphic prepotential to be

$$\Im F\left(\frac{\phi^\Lambda + ip^\Lambda}{2}\right) = \frac{p^1\phi^0 - \phi^1p^0}{(\phi^0)^2 + (p^0)^2} - \frac{12}{\pi} \Re \sum_{n=1}^{\infty} \log(1 - q^n) - \frac{(\phi^1\phi^0 + p^1p^0)B + (p^1\phi^0 - \phi^1p^0)A}{4[(\phi^0)^2 + (p^0)^2]}.$$

We have that for  $\mathcal{N} = 4$ ,  $\Lambda = 0, \dots, 23$  (in 3.1), the Renormalised action is given by

$$\mathcal{S}_{\text{Ren}} = \frac{\pi q_\Lambda \phi^\Lambda}{c} - \frac{4\pi}{c} \Im F\left(\frac{\phi^\Lambda + ip^\Lambda}{2}\right).$$

Explicitly,

$$\mathcal{S}_{\text{Ren}} = \frac{\pi q_\Lambda \phi^\Lambda}{c} + \frac{4\pi (\phi^1p^0 - p^1\phi^0)}{c[(\phi^0)^2 + (p^0)^2]} - \frac{48}{c} \Re \sum_{n=1}^{\infty} \log(1 - q^n) - \frac{\pi (\phi^1\phi^0 + p^1p^0)B + (p^1\phi^0 - \phi^1p^0)A}{c[(\phi^0)^2 + (p^0)^2]}.$$

### 3.1 The One-Loop Determinant

"N=4 the fluctuation spectrum around the localizing saddle comes in complete supersymmetry pairs, so the bosonic and fermionic determinants cancel mode by mode."

$$Z_{1 \text{ loop}} = \sqrt{\frac{\det K_f}{\det K_b}} \quad (3.7)$$

### 3.2 The Instanton Contribution

Show why they are 1

### 3.3 The Contour of Integration for the Single-Centered Contribution

The attractor contour is

$$\mathcal{M}_Q = \int_{C_*} d\tau d\sigma dz e^{-i\pi \text{Tr}(\Sigma\Lambda)} Z(\Sigma) \quad (3.8)$$

with

$$C_* : \quad \Im \tilde{\Sigma} = \varepsilon^{-1} \Lambda, \quad 0 \leq \Re \tau, \Re \sigma, \Re z < 1, \varepsilon \rightarrow 0^+ \quad (3.9)$$

Using

$$\Lambda = \begin{bmatrix} 2n & \ell \\ \ell & 2m \end{bmatrix} \quad \tilde{\Sigma} = \begin{bmatrix} \sigma & -z \\ -z & \tau \end{bmatrix} \quad (3.10)$$

Thus the bounds are

$$0 \leq \tau_1 < 0 \quad 0 \leq \sigma_1 < 1 \quad 0 \leq z_1 < 1 \quad (3.11)$$

with fixed imaginary parts

$$\Im \tau = \frac{2m}{\varepsilon} \quad \Im \sigma = \frac{2n}{\varepsilon} \quad \Im z = \frac{-\ell}{\varepsilon} \quad \varepsilon \rightarrow 0^+ \quad (3.12)$$

### 3.4 The Single-Centered Contribution

$$Z = \int_{\mathcal{M}_Q} d\mu_Q \exp \left[ \frac{\pi q_\Lambda \phi^\Lambda}{c} + \frac{4\pi (\phi^1 p^0 - p^1 \phi^0)}{c [(\phi^0)^2 + (p^0)^2]} - \frac{48}{c} \Re \sum_{n=1}^{\infty} \log(1 - q^n) - \frac{\pi (\phi^1 \phi^0 + p^1 p^0) B + (p^1 \phi^0 - \phi^1 p^0) A}{c [(\phi^0)^2 + (p^0)^2]} \right].$$

# 4 Method

The Field content of  $\mathcal{N} = 4$  supergravity is give by

- 1 gravition multiplet
- 2 gravitino multiplets
- 23 vector multiplets
- 22 hyper-multiplets

## 4.1 Scalar Manifold

In 4d  $\mathcal{N} = 4$  supergravity the classica scalar manifold is given by

$$\frac{SL(2, \mathbb{R})}{SO(2)} \times \frac{SO(6, n)}{SO(6) \times SO(n)} \quad (4.1)$$

where  $n$  is the number of vector multiplets in the theory.

In a specific string compactification  $n$  is fixed by the massless spectrum after compactification.

The object behind the 22 is the Narain compactification data, equivalently the even self dual lattice  $\Gamma^{6,22}$  whose automorphism group is  $O(6, 22; \mathbb{Z})$ .

That is why the moduli space has an  $SO(6, 22)$  factor. The 22 is the number of left moving internal gauge directions that survive the heterotic compactification.

The heterotic massless fields in 10 d are

$$G_{MN} \quad B_{MN} \quad \Phi \quad A_M^A (A = 1, \dots, 16) \quad (4.2)$$

compactified on  $T^6$  the 4 vectors come from

$$G_{\mu i} (6) \quad B_{\mu i} (6) \quad A_\mu^A (16) \quad (4.3)$$

Thus, there are  $6 + 6 + 16 = 28$  abelian gauge fields. But an  $\mathcal{N} = 4$  gravity multiplet always contains 6 vectors. So, the remaining  $28 - 6 = 22$  must sit in 22 matter vector multiplets.

## 5 Definitions

**Mock-Modular form :** A mock-modular form of weight  $k$  is a holomorphic function  $f \in \mathbb{H}$ , to which there is an associated modular form  $g$  of weight  $2 - k$ , called the shadow of  $f$ , such that the sum of  $f$  and a suitable non-holomorphic integral of  $g$  transforms like a holomorphic modular form of weight  $k$ .

**Hecke Operators :** We define the operators  $\mathcal{U}_\ell, \mathcal{V}_\ell, \mathcal{T}_\ell$  ( $\ell > 0$ ) on functions  $\phi : \mathbb{H} \times \mathbb{C} \rightarrow \mathbb{C}$

$$\left( \phi \Big|_{k,m} \mathcal{U}_\ell \right) (\tau, z) = \phi(\tau, \ell z) \quad (5.1)$$

$$\left( \phi \Big|_{k,m} \mathcal{V}_\ell \right) (\tau, z) = \ell^{k-1} \sum_{\substack{\begin{bmatrix} q & b \\ c & d \end{bmatrix} \in \Gamma_1 \backslash M_2(\mathbb{Z}) \\ ad-bc=\ell}} (c\tau + d)^{-k} e^{m\ell} \left( \frac{-cz}{c\tau + d} \right) \phi \left( \frac{a\tau + b}{c\tau + d}, \frac{\ell z}{c\tau + d} \right) \quad (5.2)$$

$$\left( \phi \Big|_{k,m} \mathcal{T}_\ell \right) = \ell^{k-4} \sum_{\substack{M \in \Gamma_1 \backslash M_2(\mathbb{Z}) \\ \det M = \ell^2 \\ \text{g.c.d } M = \square}} \sum_{X \in \mathbb{Z}^2} \sum_{\mathbb{Z}^2} \phi \Big|_{k,m} M \Big|_m X \quad (5.3)$$

In using these definitions we implement the slash operator which is given by: For a matrix

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ or for a lattice transformation } \Big|_m X$$

$$\left( \phi \Big|_{k,m} M \right) (\tau, z) = (c\tau + d)^{-k} \exp \left[ -2\pi i m \frac{cz^2}{c\tau + d} \right] \phi \left( \frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d} \right). \quad (5.4)$$

**Definition 5.0.1** (Igusa cusp form exapansions).  $\phi_{10}$  is a Siegel modular form of genus 2 : a

function of a  $2 \times 2$  symmetric period matrix  $\Omega = \begin{bmatrix} \rho & \nu \\ \nu & \sigma \end{bmatrix}$

$$\frac{1}{\phi_{10}(\rho, \sigma, \nu)} = q^{-1} s^{-1} r^{-1} \prod_{(k,l,j) > 0} \left( 1 - q^k s^l r^j \right)^{-c(4kl-j^2)}$$

$$q := e^{2\pi i \rho}, s = e^{2\pi i \sigma}, r = e^{2\pi i \nu}$$

where  $c(4kl - j^2)$  are the fourier coefficients of the elliptic genus  $\chi(K3, \tau, z)$  of a K3 surface.  $\chi(K3; \tau, z)$  is a weak Jacobi form of weight 0 and index 1

$$\chi(K3; \tau, z) = \sum_{n \geq 0, j} c(4n - j^2) q^n y^j \quad (5.5)$$

There is an equivalent definition as a series expansion

$$\frac{1}{\phi_{10}(\rho, \sigma, \nu)} = \sum_{\substack{m, n, \ell \in \mathbb{Z} \\ m, n \geq -1}} g(m, n, \ell) p^m q^n y^\ell \quad m = \frac{P^2}{2}, n = \frac{Q^2}{2}, \ell = \frac{P \cdot Q}{2} \quad (5.6)$$

6 app1